

A statement consists of preconditions and conclusions.
Disproof (Refutation): Example where preconditions are true, but conclusion is false (deriving negation of conclusion)
Direct Proof
Indirect Proof (Contradiction): 1. Assumption that statement is false 2. Derive contradiction from Assumption together with precondition of the statement. This shows assumption must be false, given the preconditions of the statement $\rightarrow$ original statement must be true. If A, then $B \rightarrow$ If A, then not B.
Contrapositive: Prove If A, then B by proving If not B, then not A

Mathematical Induction: Proof of proposition P(n) for all natural numbers n with $n \geq m$:
• *basis:* proof of P(m)
• *induction hypothesis (IH):* suppose that P(k) is true for all k with $m \leq k \leq n$
• *inductive step:* proof of P(n+1) using the IH
Weak induction: IH only supposes P(k) is true for k = n.
Strong induction: Supposes P(k) is true for all $k \in \mathbb{N}_0$ with $m \leq k < n$
Structural Induction: Proof of statement for all elements of an inductively defined set
• *basis:* proof for basic elements
• *induction hypothesis (IH):* suppose that statement is true for some elements M
• *inductive step:* proof of the statement for elements constructed by applying a construction rule to M (one IS for each construction rule)

Inductive Definition: A set M can be defined inductively by specifying: basic elements, construction rules.

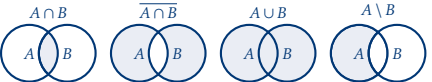
Binary Trees
• *Binary Trees:* $\square$ is a binary tree (a *leaf*), If L and R are binary trees, then $(L, \bigcirc, R)$ is a binary tree (with *inner node* $\bigcirc$)
• *Leaves of a Binary Tree:* leaves($\square$) = 1, leaves($(L, \bigcirc, R)$) = leaves(L) + leaves(R)
• *Inner Nodes of a Binary Tree:* inner($\square$) = 0, inner($(L, \bigcirc, R)$) = inner(L) + inner(R) + 1
• For all binary trees B: inner(B) = leaves(B) - 1

Sets
A set is an unordered collection of distinct objects.
Notation: *explicit* $A = \{1, 2, 3\}$, *implicit with set-builder notation* $A = \{x \mid x \in \mathbb{N}_0 \text{ and } 1 \leq x \leq 3\}$, *interval notation:* $[]$ = including, $()$ = excluding
• Natural numbers $\mathbb{N}_0 = \{0, 1, 2, \dots\}$
• Integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
• Positive Integers $\mathbb{Z}_+ = \mathbb{N}_1 = \{1, 2, \dots\}$
• Rational numbers $\mathbb{Q} = \{n/d \mid n \in \mathbb{Z}, d \in \mathbb{N}_1\}$
• Real numbers $\mathbb{R} = (-\infty, \infty)$

Set definitions
• $A \subseteq B$: A is a **subset** of B. $\subset$ for **strict subset**.
• $A \supseteq B$: A is a **superset** of B $\supset$ for **strict superset**.

Power Set P(S) is the set of all subsets of S. $P(S) = \{M \mid M \subseteq S\}$ and $\emptyset$ always in the powerset

Set operations: intersection, disjoint if $A \cap B = \emptyset$. complement, union, set difference



Commutativity of $\cup$ and $\cap$: $A \cup B = B \cup A$ and $A \cap B = B \cap A$
Associativity of $\cup$ and $\cap$: $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$

Distributivity: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

De Morgan's Law: $\overline{A \cup B} = \overline{A} \cap \overline{B}$ and $\overline{A \cap B} = \overline{A} \cup \overline{B}$
Relative Complements: $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ and $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

Cardinality

|S| of a finite set is the numbers of elements it contains.

For finite sets A and B it holds that $|A \cup B| = |A| + |B| - |A \cap B|$.

If A and B are disjoint, then $|A \cup B| = |A| + |B|$

Let S be a finite set. Then $|P(S)| = 2^{|S|}$

Bijjective Functions

- f is a function from A to B
- injective:* if $x \neq y$, then $f(x) \neq f(y)$
- surjective:* $\forall y \in B$ there is an $x \in A$ with $f(x) = y$

Cardinality not larger: Set A has cardinality less than or equal to the cardinality of set B ($|A| \leq |B|$), if there is an injective function from A to B.

Equinumerous Sets: Two sets A and B have the same cardinality ($|A| = |B|$) if there exists a bijection from A to B.

Strictly smaller cardinality: $|A| < |B|$ if there is an injective function $|A| \leq |B|$ and $|A| \neq |B|$

Countable Sets if $|A| \leq |\mathbb{N}_0|$

- Let A be a countable set. Every set B with $B \subseteq A$ is countable.

Countably Infinite Sets if $|A| = |\mathbb{N}_0|$ (there exists a bijection). Countable and Infinite

- examples: even natural numbers, n^2 perfect squares, $\mathbb{Q}_+$, $\mathbb{Q}$, $\mathbb{Z}$

Arbitrary unions: Let M be a set of sets. The union $\bigcup_{S \in M}$ is the set with $x \in \bigcup_{S \in M} S$ with $x \in S$.

- Let M be a countable set of countable sets. Then $\bigcup_{S \in M}$ is countable.
- The set $B = \{b \mid b \text{ is a binary tree} \}$ is countable

Cantor's Theorem: For every set S it holds that $|S| < |P(S)|$. Infinite Sets can have different cardinalities

Tuples and the Cartesian Product

Tuple: ordered sequence of objects, objects may occur multiple times. $(o_1, o_2, \dots, o_k)$ or $(o_1, o_2, \dots, o_k)$

The **Cartesian Product** $S_1 \times \dots \times S_n$ of sets $S_1, \dots, S_n$ is the following set of n-tuples: $S_1 \times \dots \times S_n = \{(x_1, \dots, x_n) \mid x_1 \in S_1, x_2 \in S_2, \dots, x_n \in S_n\}$

The **k-ary Catesian Power** of a set S (with $k \in \mathbb{N}_1$) is the set $S^k = \{(o_1, \dots, o_k) \mid o_i \in S \text{ for all } i \in \{1, \dots, k\}\} = S \times \dots \times S$ (k times)

- The cartesian product is not commutative ($A \times B \neq B \times A$) and not associative
- $A \times \emptyset = \emptyset \times A = \emptyset$

Relations

Let $S_1, \dots, S_n$ be sets. A relation over $S_1, \dots, S_n$ is a set $R \subseteq S_1 \times \dots \times S_n$. The arity of R is n (set of n-tuples).
Binary Relation: Relation over two sets A and B.

$\leq = \{(x, y) \mid x, y \in \mathbb{N}_0 \text{ and } x < y \text{ or } x = y\}$. We write $(x, y) \leq$ or $x \leq y$.

Homogeneous Relation: relation over set $S, R \subseteq S \times S$

Relation Properties

- Reflexivity:** for all $a \in A$ it holds that $(a, a) \in R$
- Irreflexivity:** for all $a \in A$ it holds that $(a, a) \notin R$ both at the same time only for $\emptyset$. but can be neither.
- Symmetry:** for all $a, b \in A$ it holds that $(a, b) \in R$ iff $(b, a) \in R$.
- Asymmetry:** for all $a, b \in A$ it holds that $(a, b) \in R$ then $(b, a) \notin R$.
- Antisymmetry:** for all $a, b \in A$ with $a \neq b$ it holds that $(a, b) \in R$ then $(b, a) \notin R$.
- Transitivity:** for all $a, b, c \in A$ it holds that $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$.

Types of Relations

- Equivalence Relation** $\sim$: reflexive, symmetric, transitive
Equivalence Classes: For any $x \in S$, the equivalence class of x is the set $[x]_{\sim} = \{y \in S \mid x \sim y\}$. Set of all equivalence classes $E = \{[x]_{\sim} \mid x \in S\}$. Every element of S is in some equivalence class in E. Every element of S is in at most one equivalence class in E.
- Partial Order** $\leq$: reflexive, antisymmetric, transitive.
poset: partially ordered set (S, R) where S is a set and R is a partial order over S.
least element: for all $y \in S$ it holds that $x \leq y$.
greatest element: for all $y \in S$ it holds that $y \leq x$.
minimal elements: $\nexists y \in S$ with $y \leq x$ and $x \neq y$.
maximal elements: $\nexists y \in S$ with $x \leq y$ and $x \neq y$.
- Total Order:** Relation that is total and a partial order
Total Relation: for all $x, y \in S$ at least one of xRy or yRx is true.
- Strict Order** $<$: irreflexive, asymmetric, transitive

Strict Total Order: trichotomous and a strict order (ex: $<$ over $\mathbb{N}_0$), completely ranks the elements.
Trichotomy: for all $x, y \in S$ exactly one of xRy , yRx or $x = y$ is true.

special elements: like partial order, but y where $y \neq x$

Operations on Relations

- union, intersection
- Complementary Relation:** $\overline{R} = (S_1 \times \dots \times S_n) \setminus R$
- Relation $R \subseteq A \times B$
Inverse Relation: $R^{-1} \subseteq B \times A$ given by $R^{-1} = \{(b, a) \mid (a, b) \in R\}$.
Relations $R_1 \subseteq A \times B$ and $R_2 \subseteq B \times C$.
Composition $R_2 \circ R_1 = \{(a, c) \mid \text{there is a } b \in B \text{ with } (a, b) \in R_1 \text{ and } (b, c) \in R_2\}$. Composition of Relations is **not commutative**, but **associative**: Let $S_1, \dots, S_4$ be sets and R_1, R_2, R_3 relations with $R_i \subseteq S_i \times S_{i+1}$. Then $R_3 \circ (R_2 \circ R_1) = (R_3 \circ R_2) \circ R_1$.
• **transitive closure** R^+ of R is the smallest relation over S that is transitive and has R as a subset.
• **reflexive transitive closure** R^* of R ist the smallest relation over S that is reflexive, transitive and has R as a subset. **the (r) t. closure always exists. WHY?**
• **n-fold Composition:** $R^1 = R$
 $R^0 = \{(x, x) \mid x \in S\}$ (**Identity Relation**)
 $R^i = R \circ R^{i-1} \forall i > 1$ OR $R^{n+1} = R \circ R^n$
Let R be a relation over set S. Then $R^+ = \bigcup_{i=1}^{\infty} R^i$ and $R^* = \bigcup_{i=0}^{\infty} R^i$

Functions

Notation: *explicit:* $f(a) = 4, f(b) = 2$ or $f = \{a \mapsto 4, b \mapsto 2\}$, *formula:* $f(x) = x^2 + 1$, *recurrence:* $0! = 1$ and $n! = n(n-1)$ for $n > 0$ (in programming: recursion), in terms of other functions: *inverse, composition*

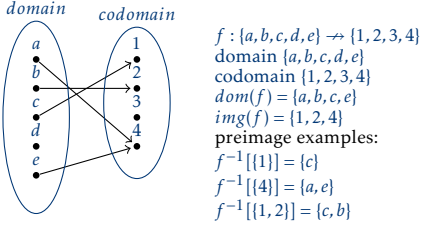
In principle, functions are just a **special kind of relations**. $f : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ with $f(x) = x^2$ or relation R over $\mathbb{N}_0$ with $R?[(x, x^2) \mid x \in \mathbb{N}_0]$. (input, output).
Functional Relations: A binary relation R over sets A and B is functional if for every $a \in A$ there is at most one $b \in B$ with $(a, b) \in R$.

Partial Functions: A partial function f from set A to set B (written $f : A \rightarrow B$) is given by a functional relation G over A and B. Relation G is called the *graph* of f. We write $f(x) = y$ for $(x, y) \in G$ and say y is the *image* of x under f. If there is no $y \in B$ with $(x, y) \in G$ then $f(x)$ is undefined.
Example $r : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Q}$ with

$r(n, d) = \begin{cases} \frac{n}{d} & \text{if } d \neq 0 \\ \text{undefined} & \text{otherwise} \end{cases}$

has graph $\{((n, d), \frac{n}{d}) \mid n \in \mathbb{Z}, d \in \mathbb{Z} \setminus \{0\}\} \subseteq \mathbb{Z}^2 \times \mathbb{Q}$

Definitions: Let $f : A \rightarrow B$ be a partial function. Set A is called the **domain** of f, set B is its **codomain**.
domain of definition: $dom(f) = \{x \in A \mid \text{there is a } y \in B \text{ with } f(x) = y\}$
image/range: $img(f) = \{y \mid \text{there is an } x \in A \text{ with } f(x) = y\}$



Preimage: The preimage contains all elements of the domain that are mapped to given elements of the codomain. $f^{-1}[Y] = \{x \in A \mid f(x) \in Y\}$

Total Functions: $f : A \rightarrow B$ is a partial function from A to B such that $f(x)$ is defined for all $x \in A$. domain = domain of definition.
Restriction And Extension: Let $f : X \rightarrow Y$ be a partial function from set X to set Y and let $A \subseteq X$. The **restriction** of f to A is the partial function $f \upharpoonright_A : A \rightarrow Y$ with $f \upharpoonright_A(x) = f(x)$ for all $x \in A$: explained: restricting the domain. **extension:** Let $f : X \rightarrow Y$ be a function and A and B be sets such that $X \subseteq A$ and $Y \subseteq B$. An extension of f to A is a function $g : A \rightarrow B$ such that $f(x) = g(x)$ for all $x \in X$. explained: making the domain bigger.

Function Composition (Composition of partial functions): Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be partial functions.

The composition of f and g is $g \circ f : A \rightarrow C$ with

$(g \circ f)(x) = \begin{cases} g(f(x)) & \text{if f is defined for x} \\ & \text{and g is defined for f(x)} \\ \text{undefined} & \text{otherwise} \end{cases}$

- Corresponds to relation composition of the graphs
- If f and g are functions, their composition is a function
- not commutative ($g \circ f \neq f \circ g$)
- associative $f \circ (g \circ f) = (f \circ g) \circ f$

Injective Function: (one-to-one) if for all $x, y \in A$ with $x \neq y$ it holds that $f(x) \neq f(y)$. maps distinct elements of its domain to distinct elements of its co-domain.
• **Composition:** If $f : A \rightarrow B$ and $g : B \rightarrow C$ are injective functions then also $g \circ f$ is injective.

Surjective Function: image equal to its codomain. for all $y \in B$ there is an $x \in A$ with $f(x) = y$. maps at least one element to every element of its co-domain.

- Composition:** If $f : A \rightarrow B$ and $g : B \rightarrow C$ are surjective functions then also $g \circ f$ is surjective.

Bijjective Function: (one-to-one correspondence) injective and surjective. every element of domain $\rightarrow$ one element of co-domain and vice-versa.

- Composition:** The composition of two bijective functions is bijective.

Inverse Functions: Let $f : A \rightarrow B$ be a bijection. The inverse function of f is the function $f^{-1} : B \rightarrow A$ with $f^{-1}(y) = x$ iff $f(x) = y$.

- For all $x \in A$ it holds that $f^{-1}(f(x)) = x$
- For all $y \in B$ it holds that $f(f^{-1}(y)) = y$
- f^{-1} is a bijection from B to A
- $(f^{-1})^{-1} = f$
- Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be bijections. Then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$

Permutations: Bijection from a set to itself. Used to describe rearrangement of objects. ex:

$\sigma : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ with $\sigma(1) = 3, \sigma(2) = 1, \sigma(3) = 2$

Permutations and Composition:

Divisibility

Let $m, n \in \mathbb{Z}$. If there exists a $k \in \mathbb{Z}$ such that $mk = n$, we say that m divides n, m is a divisor of n or n is a multiple of m and write this as $m \mid n$

Linear combinations: Let a, b and d be integers. If $d \mid a$ and $d \mid b$ then for all integers x and y it holds that $d \mid xa + yb$. Some consequences:

- $d \mid a - b$ iff $d \mid b - a$ ($x=-1, y=0$)
- If $d \mid a$ and $d \mid b$ then $d \mid a + b$ and $d \mid a - b$ ($x=1, y=1$) and ($x=1, y=-1$)
- If $d \mid a$ then $d \mid -8a$ ($x=-8, y=0$)
- Multiplication and Exponentiation:** Let $a, b, c \in \mathbb{Z}$ and $n \in \mathbb{N}_{>0}$. If $a \mid b$ then $ac \mid bc$ and $a^n \mid b^n$.
- Divisibility $\mid$ over $\mathbb{N}_0$ is a **partial order**

Modular Arithmetic

Euclid's Division Lemma: For all integers a and b with $b \neq 0$ there are unique integers q and r with $a = qb + r$ and $0 \leq r < |b|$. Number a is called the **dividend**, b the divisor, q is the **quotient** and r the **remainder**.
Modulo Operation: a mod b refers to the remainder of Euclidian division.

Congruence Modulo n: For integer $n > 1$, two integers a and b are called **congruent modulo n** if $n \mid a - b$. Written: $a \equiv b \pmod{n}$. Considers integers equivalent if they have with divisor n the same remainder ($a = qn + r$ and $b = q'n + r$).

Congruence Relation: Equivalence relation compatible with operations:

if $a \equiv a' \pmod{n}$ and $b \equiv b' \pmod{n}$:

- addition: $a + a' \equiv b + b' \pmod{n}$
- subtraction $a - a' \equiv b - b' \pmod{n}$
- multiplication $aa' \equiv bb' \pmod{n}$
- translation $a + k \equiv b + k \pmod{n} \forall k \in \mathbb{Z}$
- scaling $ak \equiv bk \pmod{n} \forall k \in \mathbb{Z}$
- exponentiation $a^k \equiv b^k \pmod{n} \forall k \in \mathbb{N}_0$

Graphs

Graph: pair $G = (V, E)$, where V is the finite set of *vertices* and $E \subseteq \{(u, v) \subseteq V \mid u \neq v\}$ is the set of *edges*. Edges are sets of 2 elements.

irreflexive symmetric relation R_E over V: uR_Ev iff $u, v \in E$

Digraph: pair $G = (N, A)$, where N is the finite set of *nodes* and $A \subseteq N \times N$ is the set of *arcs*. Arcs are pairs (tuples).

arbitrary relation R_A over N: uR_Av iff $(u, v) \in A$

Graph Terminology

- u and v are **endpoints** of / **incident** to the edge $\{u, v\} \in E$
- u and v are **adjacent** if $\{u, v\} \in E$
- neighbours:** adjacent vertices $neigh(v) = \{w \in V \mid \{v, w\} \in E\}$
- degree:** number of neighbours. $deg(v) = |neigh(v)|$

Directed Graph Terminology

- u is the *tail* and v is the *head* of the arc $(u, v) \in A$
- u and v are incident to the arc $(u, v) \in A$
- u is a *predecessor* of v and v is a *successor* of u if $(u, v) \in A$. predecessors and successors are written as $pred(v) = \{u \in N \mid (u, v) \in A\}$ and $succ(v) = \{w \in N \mid (v, w) \in A\}$
- indegree / outdegree:** $indeg(v) = |pred(v)|$ and $outdeg(v) = |succ(v)|$

Induced Graph of a Directed Graph: undirected graph induced by a directed graph. $E = \{\{u, v\} \mid (u, v) \in A, u \neq v\}$. Because of self-loops and 2 arcs can become 1 edge $|E| \leq |A|$

Degree Lemma: Let (N, A) be a *directed graph*. Then $\sum_{v \in N} indeg(v) = \sum_{v \in N} outdeg(v) = |A|$. Let (V, E) be a graph. Then $\sum_{v \in V} deg(v) = 2|E|$. **Every graph has an even number of vertices with odd degree.**

Paths and Connectivity

Walk: A walk of length n in a graph (V,E) is a tuple $\langle v_0, v_1, \dots, v_n \rangle \in V^{n+1}$ so that $\{v_i, v_{i+1}\} \in E \forall 0 \leq i \leq n$. A walk of length n in a **digraph** (N, A) is a tuple $\langle v_0, v_1, \dots, v_n \rangle \in N^{n+1}$ so that $(v_i, v_{i+1}) \in A \forall 0 \leq i \leq n$. vertices may repeat, length of walk = tuple - 1

- empty walk:** length 0 (empty tour = empty path)
- Path:** nodes don't repeat. A walk with $v_i \neq v_j \forall 0 \leq i < j \leq n$.
- Tour:** walk with $v_0 = v_n$ (start = end)
- Cycle:** tour with $n \geq 1$ (**digraphs**) or $n \geq 3$ (**graphs**) and $v_i \neq v_j \forall 1 \leq i < j \leq n$. (cycle can't be empty, self-loop is a cycle (4,4))

Reachability

successor relation $S_G : (u, v) \in S_G$ iff $\{u, v\}$ is an edge / (u, v) is an arc of G. All pairs where I can go from one node to the second.
reachability relation $R_G : (u, v) \in R_G$ iff there exists a walk from u to v. R^n of the successor relation.
Reachability as Closure: $\rightarrow$ *n-fold composition*
Let G be a graph or digraph. $(u, v) \in S_G^n$ iff there exists a walk of length n from u to v. (composition with itself n-times)
 $R_G = \bigcup_{n=0}^{\infty} S_G^n$ (union of all walks).

Graphs: Reachability is an **equivalence relation** in Graphs, but not always in Digraphs (bc. of symmetry).
Connected Components: the equivalence classes of the reachability relation of a *Graph*. connected if at most 1 c.component, exception: $(\emptyset, \emptyset)$

Weakly Connected Components: the equivalence classes of the reachability relation of the *induced graph* of a *Digraph*. weakly connected if at most 1 w.c. component, exception: $(\emptyset, \emptyset)$

Mutual Reachability Relation M_G : v reachable from u and u reachable from v.
• In Graphs $M_G = R_G$ (symmetric).
• In Digraphs M_G is an equivalence relation.

Strongly Connected Components: eq. classes of M_G . at most 1 strongly C.Component, except: $(\emptyset, \emptyset)$.

Acyclicity

Forest: acyclic Graph. **DAG**: Directed Acyclic Graph
Tree: connected forest. acyclic, connected.

- there exists exactly one **unique path** from any vertex $u \in V$ to any vertex $v \in V$.
 - A **leaf** of G is a vertex $v \in V$ with $deg(v) = 1$.
 - tree with number of vertices $|V| \geq 2$ has at least 2 leaves.
 - number of edges: tree with $V \neq \emptyset$ has $|E| = |V| - 1$
 - **in forests**: C=connected components. $|E| = |V| - |C|$
- Rooted Tree**: Recursive Data Structure.

- 1. Select any vertex v. Make v the root of the tree.
- 2. Initial state: v *pending* (v added to tree but children not), no *processed*
- 3. Repeat: Select any pending vertex u, make all neighbours v of u that are not yet processed children of u and mark them as pending, change u from pending to processed.

Further Topics in Graph Theory

Subgraph: of a **graph** (V,E) is a graph (V',E') with $V' \subseteq V$ and $E' \subseteq E$. analogously for **digraphs**. Defines a partial order on graphs.
Induced Subgraph: Let G = (V, E) be a **graph**, and let $V' \subseteq V$. The subgraph of G induced by V' is the graph (V', E') with $E' = \{\{u,v\} \in E \mid u,v \in V'\}$.
Let G = (N,A) be a **digraph**, and let $N' \subseteq N$. The subgraph of G induced by N' is the digraph (N',A') with $A' = \{\{u,v\} \in A \mid u,v \in N'\}$.

- Induced Subgraphs are the largest subgraphs.
- Subgraphs induced by the connected components of a forest are trees.
- **Counting Subgraphs**: No. of Subgraphs not easy to answer. No. of Induced Subgraphs: $2^{|V|}$

Isomorphism: graphs G = (V,E) and G' = (V',E').
An isomorphism from G to G' is a bijective function $\sigma : V \rightarrow V'$ such that for all $u,v \in V : \{u,v\} \in E$ iff $\{\sigma(u), \sigma(v)\} \in E'$. Written: $G \cong G'$.

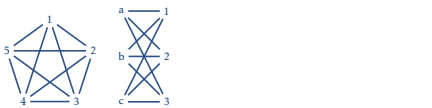
- **Isomorphisms**: Identity Function, Inverse, Composition (when defined over matching sets of vertices) $\rightarrow$ Equivalence Relation
- **Proving**: Give an example of an isomorphism
- **Disproving**: Identify *graph invariant* (property that must be the same): no. edges, no. vertices, max/min degree, number of connected component, ...

Symmetry / Automorphism: Isomorphism between Graph and itself.

Group: All symmetries of a graph.

- **Rotation**: $\sigma_1 = \{1 \mapsto 2, 2 \mapsto 3, 3 \mapsto 4, 4 \mapsto 5, 5 \mapsto 1\}$
- **Reflection**: $\sigma_1 = \{1 \mapsto 5, 2 \mapsto 4, 3 \mapsto 3, 4 \mapsto 2, 5 \mapsto 1\}$

Planarity and Minors



K_5 and $K_{3,3}$ are the smallest non-planar graphs. A graph is planar iff it does not contain K_5 or $K_{3,3}$.

Planar Embedding: Picture of a graph with no edge crossings. A graph is **planar** if such an embedding exists. All Edges straight lines.
Minor: We say that a graph G' is a minor of a graph G if it can be obtained from G through a sequence of transformations of the following kind:
• remove a vertex (of degree 0)
• remove an edge
• **Edge Contraction**: combine connected vertices u and v into a single vertex uv. neighbours of uv: $neigh(u) \cup neigh(v)$.

Wagner's Theorem: A graph is planar iff it does not contain K_5 or $K_{3,3}$ as a minor. (forbidden minors)
Minor-Hereditary Properties: If G is **planar/acyclic**, then all its minors are also planar.
• **Graph Minor Theorem**: Let Π be a minor-hereditary property of graphs. Then there exists a finite set of forbidden minors $F(\Pi)$ such that: A graph has property Π iff it does not have any graph from $F(\Pi)$ as a minor. ex: forbidden minor for acyclicity is K_3 .

Propositional Logic

Formal Logic: Describe what to accomplish, not how to accomplish it.

Propositional Logic: logic without numbers or objects.

- **propositions**: statements that can be true or false.
- **atomic propositions**: cannot be split
- **logical connectives**: and, or, negation, if...then

Syntax: Let A be a set of atomic propositions. The set of propositional formulas over A is defined as follows:

- Every **atom** $a \in A$ is a prop. formula over A
 - If φ is a p. formula, then so is its **negation** $\neg\varphi$.
 - If φ and ψ , then so is the **conjunction** $(\varphi \wedge \psi)$ and
 - If φ and ψ , then so is the **disjunction** $(\varphi \vee \psi)$ or
 - **implication if...then** $(\varphi \rightarrow \psi)$ abbrev. for $(\neg\varphi \vee \psi)$
 - **biconditional** iff $(\varphi \leftrightarrow \psi) = ((\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi))$
- Semantics**: Formally define what a formula means.
truth assignment (or **interpretation**) for a set of atomic propositions A is a function $I : A \rightarrow \{0, 1\}$ (assigns a truth value to every a.p.)

A p. formula φ holds under I ($I \models \varphi$) according to:
 $I \models a$ iff $I(a) = 1$ (for $a \in A$) (true)
 $I \models \neg\varphi$ iff not $I \models \varphi$
 $I \models (\varphi \wedge \psi)$ iff $I \models \varphi$ and $I \models \psi$ (both true)
 $I \models (\varphi \vee \psi)$ iff $I \models \varphi$ or $I \models \psi$ (at least one true)

Terminology:

- $I \models \varphi$: I is a model of φ , a is true under I
- $I \not\models \varphi$: I is no model of φ
- **Metalinguage**: $\models$ and $\not\models$
- **Properties of Propositional Formulas**:
- **satisfiable** if φ has at least one model (possible)
ex: $A \wedge B$. $I = \{A \mapsto 1, B \mapsto 1\}$, proof that $I \models (A \wedge B) \rightarrow$ result in at least one row is yes
- **unsatisfiable** if φ not satisfiable
 $\rightarrow$ result in all rows is no
- **valid/tautology** φ true under every interpretation
ex: Consider all possible interpretations.
 $\rightarrow$ result in all rows is yes
- **falsifiable** if φ is not tautology (possible to be false)
ex: $A \wedge B$. $I = \{A \mapsto 0, B \mapsto 1\}$, proof that $I \not\models (A \wedge B) \rightarrow$ result in at least one row is no

Truth Tables: example for $\varphi = ((A \rightarrow B) \vee (\neg B \rightarrow A))$

I(A)	I(B)	$I \models \neg B$	$I \models (A \rightarrow B)$	$I \models (\neg B \rightarrow A)$	$I \models \varphi$
0	0	Yes	Yes	No	Yes
0	1	No	Yes	Yes	Yes
1	0	Yes	No	Yes	Yes
1	1	No	Yes	Yes	Yes

Logical Equivalence: Two propositional formulas φ and ψ are (logically) equivalent ($\varphi \equiv \psi$) if for all interpretations I for A it is true that $I \models \varphi$ iff $I \models \psi$.
• **associativity** $((\varphi \vee \psi) \vee \chi) \equiv (\varphi \vee (\psi \vee \chi))$ (sf $\wedge$)
• **idempotence** $(\varphi \vee \varphi) \equiv \varphi$ (same for $\wedge$)
• **commutativity** $(\varphi \vee \psi) \equiv (\psi \vee \varphi)$ (same for $\wedge$)
• **absorption**: $(\varphi \wedge (\varphi \vee \psi)) \equiv \varphi$ and $(\varphi \vee (\varphi \wedge \psi)) \equiv \varphi$
• **distributivity**: $(\varphi \wedge (\psi \vee \chi)) \equiv ((\varphi \wedge \psi) \vee (\varphi \wedge \chi))$
 $(\varphi \vee (\psi \wedge \chi)) \equiv ((\varphi \vee \psi) \wedge (\varphi \vee \chi))$

- **double negation**: $\neg\neg\varphi \equiv \varphi$
- **De Morgan**: $\neg(\varphi \wedge \psi) \equiv (\neg\varphi \vee \neg\psi)$ and $\neg(\varphi \vee \psi) \equiv (\neg\varphi \wedge \neg\psi)$
- **Tautology Rules**: $(\varphi \vee \psi) \equiv \varphi$ if φ tautology
 $(\varphi \wedge \psi) \equiv \psi$ if φ tautology
- **Unsatisfiability**: $(\varphi \vee \psi) \equiv \psi$ if φ unsatisfiable
 $(\varphi \wedge \psi) \equiv \varphi$ if φ unsatisfiable

Substitution Theorem: Let φ and φ' be equivalent propositional formulas over A. Let ψ be a propositional formula with (at least) one occurrence of the subformula φ . Then ψ is equivalent to ψ' , where ψ' is constructed from ψ by replacing an occurrence of φ in ψ with φ'

Simplified Notations

Implicit Placement of Parentheses
 $\neg$ stronger $\wedge$ stronger $\vee$ stronger $\rightarrow$ or $\leftrightarrow$
 $A \vee \neg C \wedge B \rightarrow A \vee \neg D = (A \vee (\neg C \wedge B)) \rightarrow (A \vee \neg D)$
Short Notations for Conjunctions and Disjunctions
 $\bigwedge_{i=1}^n \varphi_i = (\varphi_1 \wedge \varphi_2 \wedge \dots \wedge \varphi_n)$
 $\bigvee_{\varphi \in X} \varphi = (\varphi_1 \vee \varphi_2 \vee \dots \vee \varphi_n)$ for $X = \{\varphi_1, \varphi_2, \dots, \varphi_n\}$
analogous for $\vee$
Corner Cases: Empty Set, Only one Element in Set
 $\bigwedge_{\varphi \in \emptyset} \varphi$ is a tautology
 $\bigvee_{\varphi \in \emptyset} \varphi$ is unsatisfiable.
 $\bigwedge_{\varphi \in \{x\}} \varphi = \bigvee_{\varphi \in \{x\}} \varphi = x$

Normal Forms

union $\cup$, **OR**, **disjunction** $\vee$, **clause**
intersection $\cap$, **AND**, **conjunction** $\wedge$, **monomial**
A representation of a formula with syntactic restrictions.
literal: atomic proposition or negation $A, \neg A$
clause: disjunction of literals. $(Q \vee \neg P \vee \neg S \vee R)$
monomial: conjunction of literals. $(Q \wedge \neg P \wedge \neg S \wedge R)$
clause and monomial also used for corner case with only one literal $A, \neg A$

Conjunctive Normal Form: clause $\wedge$ clause
conjunction of clauses. form: $\bigwedge_{i=1}^n \bigvee_{j=1}^{m_i} L_{ij}$

with $n, m_i > 0$ (for $1 \leq i \leq n$) where the L_{ij} are literals.
ex: $((\neg P \vee Q) \wedge R \wedge (P \vee \neg S))$

Disjunctive Normal Form: monomial $\vee$ monomial
disjunction of monomials. form: $\bigvee_{i=1}^n \bigwedge_{j=1}^{m_i} L_{ij}$

with $n, m_i > 0$ (for $1 \leq i \leq n$) where the L_{ij} are literals.
ex: $((\neg P \wedge Q) \vee R \vee (P \wedge \neg S))$

a single literal is in both DNF and CNF.

Construction of Normal Form:

- Replace abbreviations $\rightarrow$ and $\leftrightarrow$ by definitions
 $\rightarrow$ formula structure only $\vee, \wedge, \neg$
- Move negations inside: **De morgan**, **Double Neg.**
 $\rightarrow$ formula structure only $\vee, \wedge$, literals
- Distribute $\vee$ over $\wedge$ with **distributivity**
 $\rightarrow$ formula structure: CNF (vice-versa for DNF)

Existence of Equivalent Formula in NF: For every formula φ there is at least 1 logically equivalent formula in CNF & DNF.
Tautology: A formula in CNF is a tautology iff every clause is a tautology.

Satisfiable: A formula in DNF is satisfiable iff at least one monomial is satisfiable.

Knowledge Bases

- A **Knowledge Base** over A is a set of propositional formulas over A. A truth assignment I for A is a model for KB ($I \models KB$) if I is a model for every formula $\varphi \in KB$.
- **satisfiable**: is KB has at least one model (true for all elements in the KB) **unsatisfiable**: not satisfiable
 - **valid**: every interpretation is a model for KB
 - **falsifiable**: find model which is false for at least one element in KB

Logical Languages

Let KB be a set of formulas and φ a formula. We say that KB logically implies φ . Written $KB \models \varphi$ if all models of KB are also models of φ .

$\models$ has multiple meanings:

$KB \models \varphi$:

$I \models \varphi$:

Theorems:

Deduction: $KB \cup \{\varphi\} \models \psi$ iff $KB \models (\varphi \rightarrow \psi)$

Contraposition: $KB \cup \{\varphi\} \models \neg \psi$ iff $KB \cup \{\psi\} \models \neg \varphi$

Contradiction: $KB \cup \{\varphi\}$ unsatisfiable iff $KB \models \neg \varphi$

Inference

Syntactic Inference Rules give us a mechanical method to produce formulas that are logical consequences of given formulas.

Inference Rule:

Every model of $\varphi_1, \dots, \varphi_k$ is a model of ψ $\frac{\varphi_1, \dots, \varphi_k}{\psi}$

$\varphi_1, \dots, \varphi_k$ logically implies ψ
Axiom: Inference Rule with $k = 0$.
Calculus / Proof System: Set of Inference Rules.
Inference Rules for Propositional Logic:

Modus ponens	$\frac{\varphi, (\varphi \rightarrow \psi)}{\psi}$
Modus tollens	$\frac{\neg\psi, (\varphi \rightarrow \psi)}{\neg\varphi}$
$\wedge$ -elimination	$\frac{(\varphi \wedge \psi)}{\varphi} \quad \frac{(\varphi \wedge \psi)}{\psi}$
$\wedge$ -introduction	$\frac{\varphi, \psi}{(\varphi \wedge \psi)}$
$\vee$ -introduction	$\frac{\varphi}{(\varphi \vee \psi)} \quad \frac{\psi}{(\varphi \vee \psi)}$
$\leftrightarrow$ -elimination	$\frac{\varphi \leftrightarrow \psi}{(\varphi \rightarrow \psi)} \quad \frac{\varphi \leftrightarrow \psi}{(\psi \rightarrow \varphi)}$

Derivation: A derivation or proof of a formula φ from a KB is a sequence of formulas $\psi_1, \dots, \psi_k$ with
• $\psi_k = \varphi$
• for all $i \in \{1, \dots, k\}$: $\psi_i \in KB$ or ψ_i is the result of the application of an inference rule to elements from $\{\psi_1, \dots, \psi_{i-1}\}$

Ex: Find derivation of $(S \wedge R)$ from KB. Start with KB and apply inference rules until you get $(S \wedge R)$
Correctness and Completeness: $KB \vdash_c \varphi$ means we can derive φ from a calculus C.

A calculus C is **correct** if for all KB and φ $KB \vdash_c \varphi$ implies $KB \models \varphi$
A calculus C is **complete** if for all KB and φ $KB \models \varphi$ implies $KB \vdash_c \varphi$

Predicate Logic

Signatures define allowed symbols (analogous to atoms in propositional logic) 4-tuple $S = (V, C, F, P)$

- finite or countable Set V of **variable symbols**:
- " Set C of **constant symbols**
- " Set F of **function symbols** + arity $\forall f \in F$
- " Set P of **predicate symbols** + arity $\forall P \in P$

Arity: ar(f), ar(P) = numbers of arguments. Described *k-ary* symbol = symbol s with arity k.

Terms: associated with objects by semantics (no analogy)
A **term** (over S) is inductively constructed according to the following rules:

- every variable symbol $v \in V$ is a term
- every constant symbol $c \in C$ is a term
- If $t_1, \dots, t_k$ are terms and $f \in F$ is a function symbol with arity k, then $f(t_1, \dots, t_k)$ is a term.

Formulas: associated with truth values (true, false) by semantics. (analogous to formulas in pl).

Inductive Definition:

- If $t_1, \dots, t_k$ are terms (over S) and $P \in P$ is a k-ary predicate symbol, then the **atomic formula** (**atom**) $P(t_1, \dots, t_k)$ is a formula over S.
- If t_1 and t_2 are terms (over S), then the **identity** $(t_1 = t_2)$ is a formula over S.
- If $x \in V$ is a variable symbol and φ a formula o.S, then the **universal quantification** $\forall x\varphi$ and the **existential quantification** $\exists x\varphi$ are formulas o. S.
- If φ is a formula over S, then so is its **negation** $\neg\varphi$
- If φ and ψ are formulas over S, then so are the **conjunction** $(\varphi \wedge \psi)$ and the **disjunction** $(\varphi \vee \psi)$

Abbreviations: Sequences of quantifiers: $\forall x\forall y\forall z\varphi \rightarrow \forall xyz\varphi$
Parentheses like propositional logic but $\forall$ and $\exists$ stronger

Semantics

Interpretation: Pair $I = \langle U, I \rangle$
• **Universe U**: non-empty set
• **Function** I assigns meaning to $c^I \in U$ for constant symbols $c \in C$
k-ary function symbols $f \in F$: $f^I : U^k \rightarrow U$
k-ary predicate symbols $P \in P$: $P^I \subseteq U^k$
• A **variable assignment** (for S and Universe U) is a function $\alpha : V \rightarrow U$ (assigns meaning to variables)
Example $(\forall x(\text{Block}(x) \rightarrow \text{Red}(x)) \wedge \text{Block}(a))$
"For all objects x: if x is a block, then x is red. Also, the object called a is a block."

- **Terms** are interpreted as objects.
- **Unary predicates** denote properties of objects
- **General predicates** denote relations between objects
- **Universally quantified formulas** ($\forall$) are true if they hold for every object in the universe.
- **Existentially quantified formulas** ($\exists$) are true if they hold for at least one object in the universe.

Interpretation of Terms: The interpretation of Term t under Interpretation I and variable assignment α (t^I, α is the element of the universe U, defined as follows:

- variable term $t = x$ with $x \in V$: $x^I, \alpha = \alpha(x)$
- constant term $t = c$ with $c \in C$: $c^I, \alpha = c^I$
- function term $t = f(t_1, \dots, t_k)$:

$f(t_1, \dots, t_k)^I, \alpha = f(I(t_1^I, \alpha), \dots, I(t_k^I, \alpha))$
Formula is Satisfied or True: I and α satisfy a predicate logic formula φ (φ is true under I and α): $I, \alpha \models \varphi$, according to these inductive rules:

- $I, \alpha \models P(t_1, \dots, t_k)$ iff $(t_1^I, \alpha), \dots, (t_k^I, \alpha) \in P^I$
- $I, \alpha \models (t_1, \dots, t_k)$ iff $t_1^I, \alpha = t_2^I, \alpha$
- $I, \alpha \models \neg\varphi$ iff $I, \alpha \not\models \varphi$
- $I, \alpha \models (\varphi \wedge \psi)$ iff $I, \alpha \models \varphi$ and $I, \alpha \models \psi$
- $I, \alpha \models (\varphi \vee \psi)$ iff $I, \alpha \models \varphi$ or $I, \alpha \models \psi$
- $I, \alpha \models \forall x\varphi$ iff $I, \alpha[x := u] \models \varphi$ for all $u \in U$
- $I, \alpha \models \exists x\varphi$ iff $I, \alpha[x := u] \models \varphi$ for at least one $u \in U$

where $\alpha[x := u](z) = \begin{cases} u & \text{if } z = x \\ \alpha(z) & \text{if } z \neq x \end{cases}$

Free and Bound Variables
Bound Variables: Occurrences of Variables that are bound by a surrounding quantifier.

Variables of a Term: Let t be a term. The set of variables that occur in t (**var(t)**)

- $\text{var}(x) = \{x\}$ for variable symbols x
- $\text{var}(c) = \emptyset$ for constant symbols c
- function terms: $\text{var}(f(t_1, \dots, t_k)) = \text{var}(t_1) \cup \dots \cup \text{var}(t_k)$

Ground Term: Term t with $\text{var}(t) = \emptyset$
ex: $\text{var}(\text{product}(x, \text{sum}(k, y))) = \text{var}(x) \cup \text{var}(\text{sum}(k, y)) = \{x\} \cup \text{var}(k) \cup \text{var}(y) = \{x\} \cup \emptyset \cup \{y\} = \{x, y\}$

Free Variables: (value of the variable assignment matters)
The set of free variables of predicate logic formula φ (**free(φ)**):

- $\text{free}(P(t_1, \dots, t_k)) = \text{var}(t_1) \cup \dots \cup \text{var}(t_k)$
- $\text{free}(t_1 = t_2) = \text{var}(t_1) \cup \text{var}(t_2)$
- $\text{free}(\neg\varphi) = \text{free}(\varphi)$
- $\text{free}((\varphi \wedge \psi)) = \text{free}((\varphi \vee \psi)) = \text{free}(\varphi) \cup \text{free}(\psi)$
- $\text{free}(\forall x\varphi) = \text{free}(\exists x\varphi) = \text{free}(\varphi) \setminus \{x\}$

Let φ be a formula and α and β variable assignments with $\alpha(x) = \beta(x) \forall$ free variables x of φ . Then $I, \alpha \models \varphi$ iff $I, \beta \models \varphi$.

Closed Formulas/Sentences: Formula φ without free variables ($\text{free}(\varphi) = \emptyset$). If φ is a **sentence**, we can write $I \models \varphi$ instead of $I, \alpha \models \varphi$ (variable assignment α irrelevant)

Ground Formula: Closed formula without quantifiers.

Terminology for Formulas: $\rightarrow$ prop. logic
Satisfied/True Sets of Formulas: Let S be a signature, Φ a set of formulas over S, I an interpretation for S and α a variable assignment for S and the universe of I. I, α satisfy formulas / Φ is true under I and α ($I, \alpha \models \Phi$) if $I, \alpha \models \varphi \forall \varphi \in \Phi$.

Logical Equivalence and Consequence: $\rightarrow$ prop. logic
 $(\forall x\varphi \wedge \forall x\psi) \equiv \forall x(\varphi \wedge \psi)$
 $(\forall x\varphi \vee \forall x\psi) \equiv \forall x(\varphi \vee \psi)$ but not the converse
 $\exists x(\varphi \vee \psi) \equiv (\exists x\varphi \vee \exists x\psi)$
 $\exists x(\varphi \wedge \psi) \equiv (\exists x\varphi \wedge \exists x\psi)$ but not the converse
 $(\forall x\varphi \wedge \psi) \equiv \forall x(\varphi \wedge \psi)$ and $(\exists x\varphi \wedge \psi) \equiv \exists x(\varphi \wedge \psi)$ if $x \notin \text{free}(\psi)$
 $(\forall x\varphi \vee \psi) \equiv \forall x(\varphi \ve$